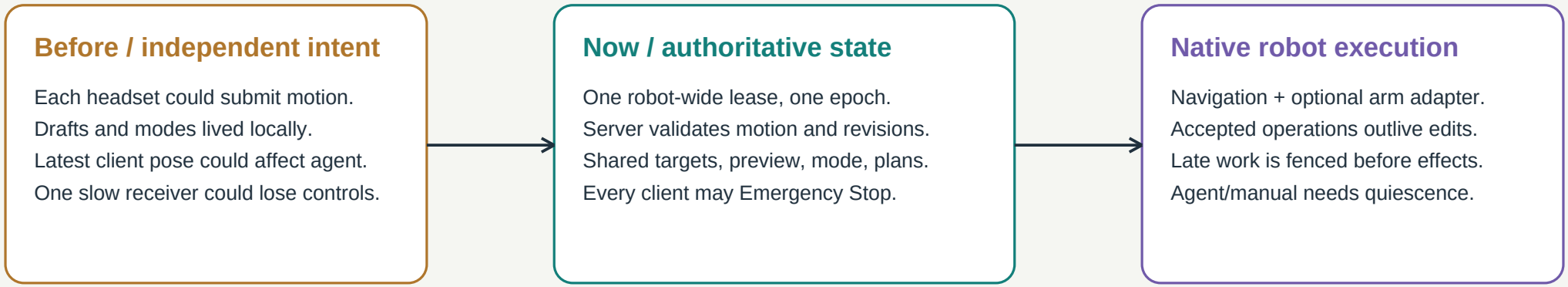
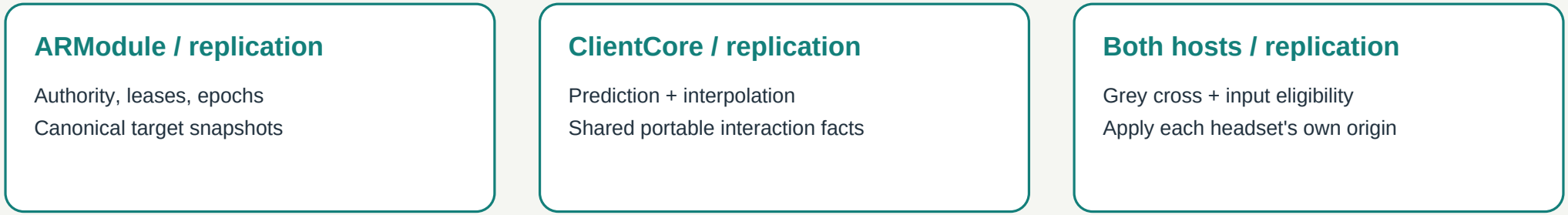
Multiple connections need one robot owner.

The original transport accepted multiple clients; shared interaction authority was incomplete.



Explicit replication boundaries



Scope: one robot, mixed Specs/WebXR peers; no upstream DimOS edits or scene-file edits.

Fast for the owner. Smooth for observers.

Local prediction is immediate; the server remains the source of accepted robot state.

Client A / dragging

Render local pose immediately.
Acquire / renew robot-wide lease.
Send fenced target samples.

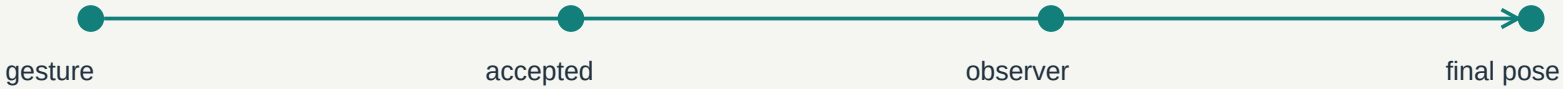
ARModule / canonical odom

Validate epoch + lease + sequence.
Publish shared state at up to 20 Hz.
Keep nav and arm groups distinct.

Client B / observing

Buffer samples for about 100 ms.
Interpolate position + rotation.
Render using B's tracking origin.

Replica timing



Owner prediction never waits for the network. Native commands wait for authority.

Observers do not extrapolate. Quaternion interpolation takes the shortest rotation path.
Final samples settle exactly; stale / reordered snapshots cannot revive an old gesture.
Reliable controls preserve order. Coalesced telemetry avoids unbounded backlog; failed peers retire.
Late join and reconnect rebuild from full state; parked targets remain available for handover.

Only one client plans the selected target.

The reservation survives the first release so the two-gesture preview workflow stays exclusive.

1 / First gesture

A grabs valid floor target.
Robot-wide lease granted.

2 / First release

Shared preview remains.
A keeps planning reservation.

3 / Second gesture

A sets heading and confirms.
Server accepts native goal.

4 / Handover

Reservation ends.
B can grab shared target.

Observer B / unavailable

Sees A's target and advisory route.
Grey marker with cross; grabbing is disabled.
Emergency Stop and cancellation controls stay reachable.

Reservation cleanup

Confirm / cancel / deselect / tracking loss.
Disconnect or lease expiry also retires authority.
Normal live release opens authority and keeps the target.

A 500 ms lease is renewed during the reservation. Losing renewal cannot leave an immortal lock.
Arm execution has a separate robot-owned busy state; ending an edit does not unlock an active operation.

One world, independently aligned headsets.

AprilTag math is reusable across clients; calibration and capture timing decide physical accuracy.

Robot side

Native odometry
Measured base-to-tag mount

Observed tag

Known black-square size
Camera optical pose at exposure

Headset side

Per-client tracking origin
Never shared between headsets

T_odom_client = (T_odom_base x T_base_tag) x inverse(T_client_camera x T_camera_tag)

Implemented corrections

Reject Specs pose brackets farther than 100 ms.
Bound deliberate realignment capture recovery.
Keep canonical targets in odom; reproject on correction.
Reference resets fence old commands and captures.

Physical checks still required

Verify tag size and rigid robot mount by measurement.
Validate camera intrinsics, crop and extrinsics per device.
Check two-headset target overlap while both move.
Wrong size or mount can still give low reprojection error.

Synthetic tag/PnP checks and unit tests support the transform chain; they do not certify device alignment.
Built-in Quest calibration assumptions still need measured-device verification.

Test the races, then verify on devices.

Automated coverage exercises shared state and native boundaries; hardware remains a separate acceptance step.

Unit / deterministic clocks

Competing grabs; delayed grants.
Lease expiry; renew; cancellation.
Epoch / sequence / revision fences.
Predicted and interpolated targets.
Native plan / execute late results.
Exposure timestamps and alignment.

Integration / production owners

Real multi-socket ARModule tests.
Two browser contexts, one server.
Shared preview and grey cross.
Confirm / handover / disconnect.
Native ports simulated explicitly.
Existing browser suite retained.

Device / acceptance

Specs + Quest concurrent session.
Moving tags, occlusion, relocalize.
Wi-Fi jitter and temporary loss.
Observed drag smoothness.
Physical stop / backend feedback.
Actual robot and camera mounts.

Repository validation

Acceptance gate: ./launcher/scripts/run-ci.sh. Device acceptance remains pending.

Native agent turns cannot be forcibly cancelled; a running turn may issue later skills after a software stop.
Manual admission waits for confirmed turn/navigation/following quiescence. Validate hardware stop behavior.
Native goal-monitor races can still stop a newer goal; source-correlated transport cannot repair that algorithm.
No finite test suite proves every scenario. Physical safety, alignment and visual latency need device evidence.

Implementation map

dimos/ar/replication | ClientCore/replication | both host replication folders | PROTOCOL.md